

EEG Channel Reduction

Keerat Kaur

Abstract

Electroencephalography (EEG)-based motor imagery brain-computer interfaces (BCIs) enable the decoding of imagined movements from neural activity, offering promising applications in assistive technology and neurorehabilitation. However, most BCI systems rely on high-density electrode arrays, limiting their practicality in real-world settings due to setup complexity and user discomfort. This project investigates whether accurate motor imagery classification can be achieved using reduced EEG channel configurations. Using the BCI Competition IV Dataset 2a, I classify left-hand versus right-hand motor imagery using Common Spatial Patterns (CSP) and multiple machine learning classifiers. I systematically evaluate classification performance across progressively smaller electrode subsets and quantify channel importance using a leave-one-channel-out analysis. Results demonstrate that while full-channel configurations provide the highest accuracy, reduced sensorimotor channel sets maintain competitive performance. These findings highlight a critical trade-off between electrode count and decoding performance and support the feasibility of low-electrode wearable BCI systems. Notably, classification performance remained stable despite substantial reductions in electrode count, indicating that accurate decoding can be achieved with a minimal sensor configuration.

Background & Prior Work

Motor imagery classification has been a central problem in brain-computer interface research, particularly for non-invasive EEG-based systems. When individuals imagine movements, characteristic patterns emerge in sensorimotor rhythms, especially in the mu (8–13 Hz) and beta (13–30 Hz) frequency bands. These changes, often referred to as event-related desynchronization (ERD), are most prominent over the motor cortex and can be detected using EEG. The BCI Competition IV Dataset 2a has become a standard benchmark for evaluating motor imagery classification methods (Blankertz et al., 2008). Traditional approaches frequently rely on Common Spatial Patterns (CSP), a spatial filtering technique that extracts features by maximizing variance differences between classes. CSP is commonly paired with linear classifiers such as Linear Discriminant Analysis (LDA) and Support Vector Machines (SVM), which have demonstrated strong and consistent performance across subjects. More recently, deep learning approaches such as convolutional neural networks and transformer-based models have been explored for EEG decoding (Schirrneister et al., 2017). While these models can capture complex nonlinear relationships, they often require large datasets and may lack interpretability compared to traditional methods. Despite extensive work on improving classification accuracy, relatively little attention has been given to the practical constraints of electrode count. High-density EEG systems are difficult to deploy outside laboratory settings due to long setup times and user discomfort. As a result, there is increasing interest in determining whether reliable motor imagery decoding can be achieved using fewer electrodes.

This project addresses this gap by systematically evaluating how classification performance changes as EEG channels are reduced and identifying the most informative electrodes for motor imagery classification.

Data Preprocessing

The EEG data used in this study were obtained from the BCI Competition IV Dataset 2a, which contains recordings from nine subjects performing motor imagery tasks involving the left hand, right hand, feet, and tongue. For this project, only left-hand and right-hand motor imagery trials were used to formulate a binary classification problem. Data preprocessing was performed using the MNE-Python library. The first 22 EEG channels were retained, and channel names were mapped to the standard 10–20 system to allow for consistent spatial interpretation. Non-EEG channels were excluded. To isolate neural activity relevant to motor imagery, a band-pass filter between 8–30 Hz was applied, capturing the mu and beta frequency bands. Event markers corresponding to motor imagery cues were extracted, and EEG signals were segmented into epochs from 0.5 to 4.0 seconds following cue onset. This time window corresponds to the period during which motor imagery-related neural activity is most prominent. The resulting epoched data were used as input for feature extraction and classification.

Model & Methods

To extract discriminative features from the EEG signals, I used Common Spatial Patterns (CSP). CSP computes spatial filters that maximize variance for one class while minimizing variance for the other, effectively emphasizing class-specific neural activity patterns. This makes CSP particularly well-suited for motor imagery classification tasks.

The CSP features were used as input to three classification models:

1. Linear Discriminant Analysis (LDA)
2. Support Vector Machine (SVM) with a linear kernel
3. Random Forest classifier

All models were implemented using scikit-learn pipelines, ensuring consistent preprocessing and feature extraction across classifiers. Model performance was evaluated using 5-fold stratified cross-validation to provide robust estimates of classification accuracy.

To investigate the effect of electrode count, multiple channel configurations were defined:

1. Full set (22 channels)
2. Sensorimotor subset (9 channels)
3. Central subset (5 channels)

4. Minimal subset (3 channels: C3, Cz, C4)

Each configuration was evaluated using the same CSP + LDA pipeline to isolate the effect of channel reduction.

To further quantify the contribution of individual electrodes, a leave-one-channel-out analysis was performed. Each channel was removed individually, and the resulting decrease in classification accuracy was recorded. Channels whose removal led to larger performance drops were considered more important for classification. Stratified cross-validation was used to ensure balanced class distributions across folds and to provide a robust estimate of model generalization performance.

Data Analysis and Results

Baseline model comparisons showed that CSP combined with LDA and SVM achieved comparable classification performance, with mean accuracies around 0.64–0.66 across cross-validation folds, while Random Forest performed slightly worse. This indicates that linear models remain highly effective for motor imagery EEG decoding, likely due to the structured nature of CSP features. As shown in Figure 1, classification accuracy remained relatively stable as the number of EEG channels was reduced. Accuracy decreased only modestly from approximately 0.66 using all 22 channels to approximately 0.63 when using only 3 channels (C3, Cz, C4), despite an over 85% reduction in electrode count. This suggests that a large proportion of the discriminative information required for motor imagery classification is concentrated in a small subset of electrodes. Importantly, the decline in performance was not uniform across channel configurations. The sensorimotor subset (9 channels) retained nearly identical performance to the full-channel configuration, indicating that channels outside motor-related regions contribute minimally to classification accuracy. This supports the hypothesis that motor imagery decoding relies primarily on localized neural activity rather than distributed signals across the entire scalp. Per-subject analysis revealed variability in classification performance across participants, with some subjects achieving consistently higher accuracy than others. This highlights the subject-dependent nature of EEG signals and suggests that individualized models or adaptive approaches may further improve performance. The leave-one-channel-out analysis (Figure 2) provided a more granular view of channel importance. Removal of electrodes over the sensorimotor cortex, particularly C3, Cz, and C4, resulted in the largest decreases in classification accuracy, confirming their critical role in motor imagery decoding. In contrast, removal of peripheral electrodes produced minimal or negligible changes in performance. This spatial pattern is further illustrated in Figure 3, which shows the topographic distribution of channel importance. Warmer regions are concentrated over the sensorimotor cortex, while other regions contribute less significantly. Together, these results demonstrate that motor imagery classification depends primarily on a localized set of electrodes, and that reducing electrode count can be achieved with minimal loss in performance.

Figure 1. Classification Accuracy Across Channel Configurations

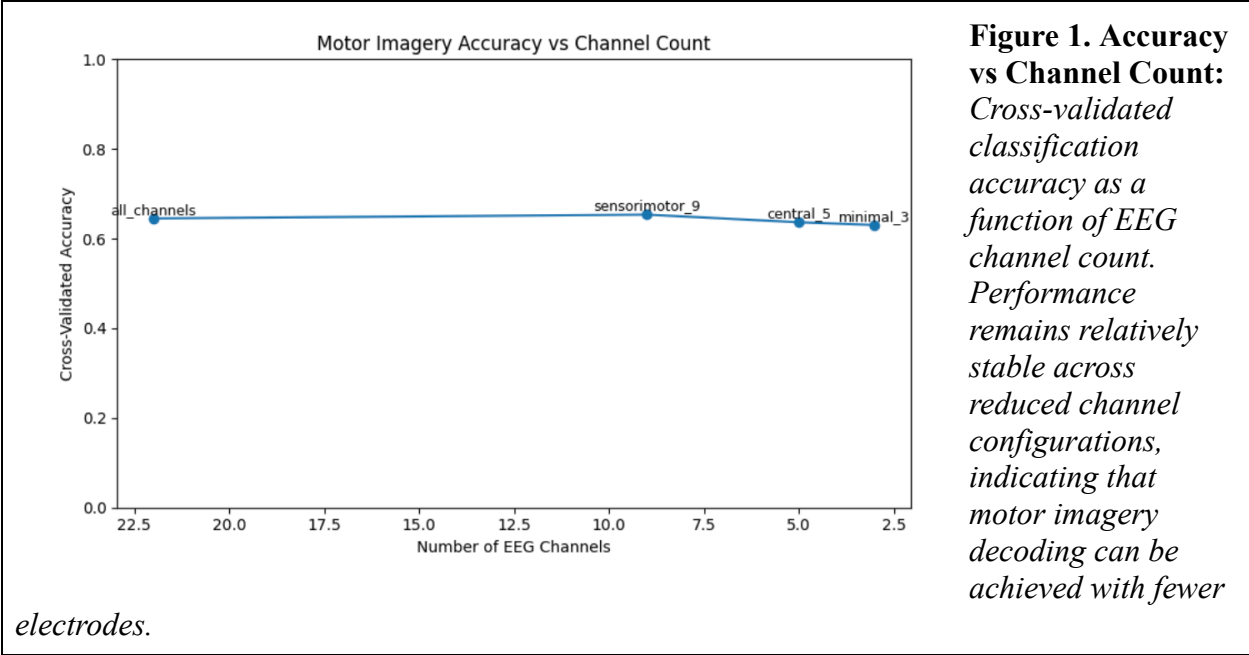


Figure 2. Leave-One-Out Channel Importance

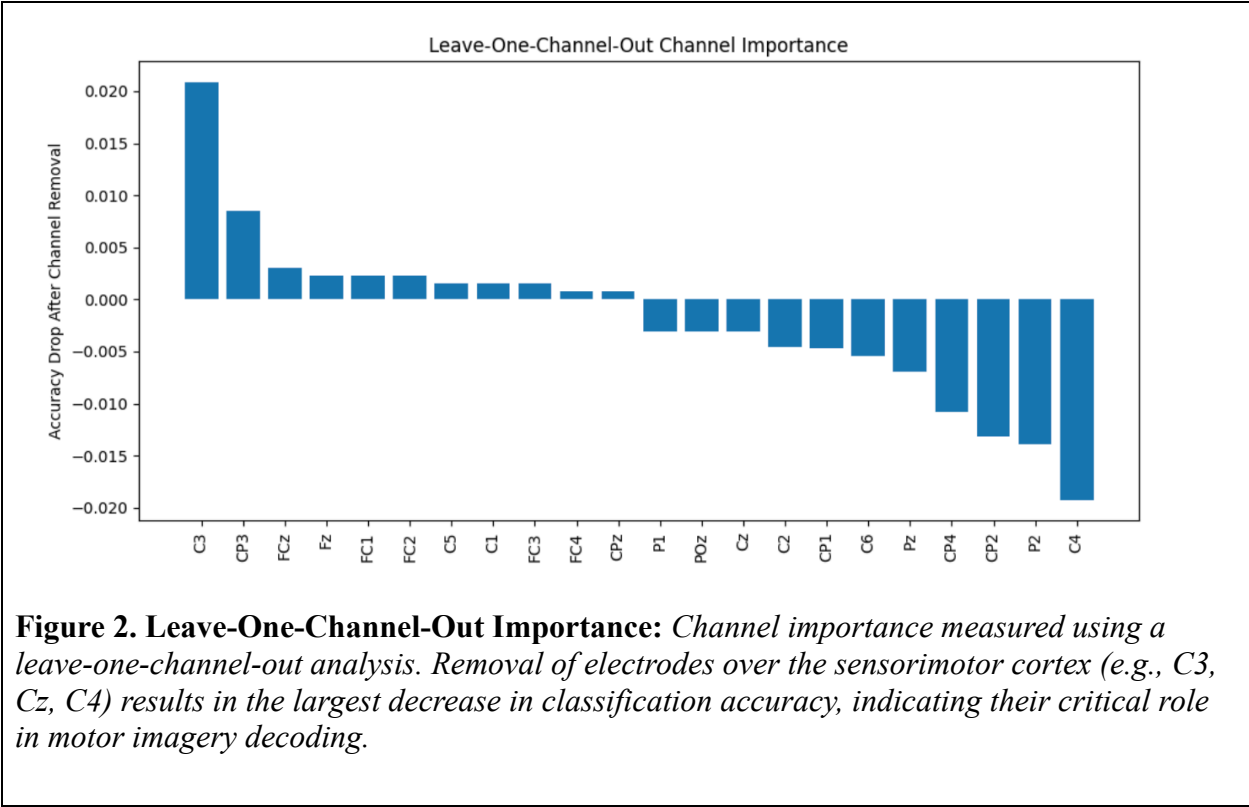
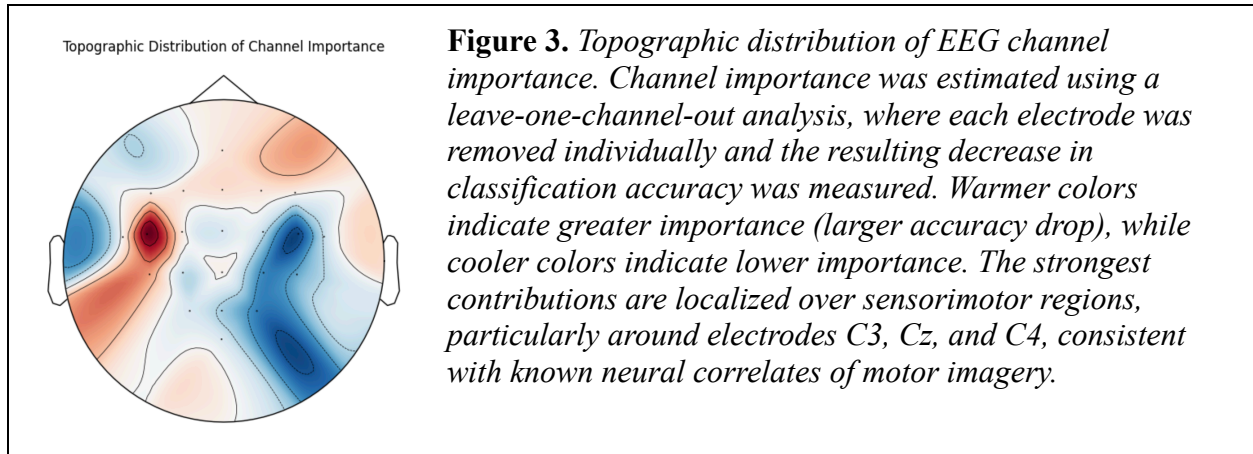


Figure 3. EEG Channel Importance Topography



Discussion

The results of this study demonstrate that motor imagery classification remains feasible even when the number of EEG electrodes is significantly reduced. While full-channel configurations provide the highest classification accuracy, reduced channel sets focused on the sensorimotor cortex retain competitive performance. This suggests that much of the relevant information for motor imagery decoding is localized to specific brain regions rather than distributed across the entire scalp. Although the identification of electrodes such as C3, Cz, and C4 as highly informative is consistent with prior literature, this finding serves as an important validation of the analysis pipeline. The alignment of model-derived importance with known neurophysiological patterns indicates that the feature extraction and classification methods are capturing meaningful neural signals rather than noise. More importantly, this project extends beyond confirming known electrode importance by quantifying the relationship between electrode count and classification performance. The results reveal a diminishing returns effect, where removing peripheral electrodes leads to relatively small decreases in accuracy, while removing central sensorimotor electrodes results in larger performance degradation. This provides practical insight into how electrode configurations can be optimized for real-world BCI systems. From a practical perspective, these findings are highly relevant for the design of wearable and real-world BCI systems. Reducing electrode count decreases setup time, improves user comfort, and lowers hardware complexity, all of which are critical for deployment outside controlled laboratory environments. The ability to maintain competitive performance with a minimal set of electrodes suggests that lightweight, portable BCI systems are not only feasible but can be effective in real-world applications.

Importantly, the reduction in classification accuracy across channel subsets was relatively modest. Even when reducing from the full electrode set to a minimal configuration of three

electrodes (C3, Cz, C4), performance remained largely stable. Further reduction to two or a single electrode resulted in more noticeable declines, but these decreases were not drastic. This suggests that motor imagery decoding is robust to channel reduction and that the majority of discriminative information is concentrated within a small subset of sensorimotor regions. The absence of a sharp performance drop indicates that simpler, low-channel configurations may be sufficient for practical BCI applications.

However, several limitations were observed within the scope of this project. Classification performance varied significantly across subjects, suggesting that motor imagery signals are highly subject-dependent. Additionally, more complex models did not consistently outperform simpler linear approaches, indicating that increased model complexity does not necessarily improve performance for EEG data. Another limitation is the absence of advanced preprocessing techniques such as artifact removal or independent component analysis, which could improve signal quality and classification performance. Future work should explore these approaches to further enhance decoding accuracy. Lastly, statistical significance testing was not performed to formally evaluate differences in classification performance across channel configurations. While observed differences in accuracy were relatively small, it is unclear whether these differences are statistically meaningful. Results may be influenced by subject variability, as EEG signals can differ substantially across individuals. The analysis also did not incorporate advanced artifact removal techniques, which may impact signal quality. The dataset used is also limited in size, and results may not fully generalize to larger or more diverse populations.

Conclusion

This project investigated whether accurate motor imagery classification can be achieved using reduced EEG electrode configurations. The results demonstrate that classification performance remains relatively stable even with substantial reductions in electrode count, with minimal performance loss observed when using a small subset of sensorimotor channels. These findings indicate that the most informative signals for motor imagery decoding are concentrated in localized brain regions rather than distributed across the entire scalp. This work provides a quantitative evaluation of the trade-off between electrode count and classification accuracy, showing that practical reductions in hardware complexity can be achieved without significantly compromising performance. This has direct implications for the design of real-world brain-computer interface systems. Overall, these results support the feasibility of developing lightweight, portable, and user-friendly BCI systems that rely on fewer electrodes, making neurotechnology more accessible for applications in rehabilitation, assistive devices, and everyday human-computer interaction.

References

- Blankertz, B., et al. (2008). The BCI competition IV: Dataset 2a.
- Schirrneister, R. T., et al. (2017). Deep learning with convolutional neural networks for EEG decoding and visualization.

